

FresVeg Agritourism & Direct Harvest Platform

Database Data Contract Summary

Official Database Data Contract Summary | August 27, 2026

Database Data Contract Summary

Technical Specification & Schema Standard for Engineering Managers

EXECUTIVE SUMMARY

This summary contract establishes the canonical data structures, table schemas, JSONB field definitions, API routes, and database constraints for the FresVeg website platform. Use this document as the reference guide for backend implementation and PostgreSQL database design.

1. Core Data Entities Summary

Table Name	Entity Domain	Business Responsibility & Contents
users	User Accounts & Roles	Stores user accounts, roles (customer, vendor, delivery), profile details, and address arrays.
farms	Agritourism Listings	Stores farm profiles, crops, fruits, livestock, stay accommodations, farm products, and photo galleries.
farm_bookings	Tour & Stay Bookings	Tracks visitor tour bookings, guest count, accommodation selections, and payment statuses.
products	Harvest Marketplace	Stores direct farm produce items with category, price, unit, and deliverability flags.
orders	Customer Orders	Tracks direct harvest purchases, JSONB items, shipping address, and delivery driver status.
shops	Vendor Farm Shops	Stores vendor shop storefront profiles associated with farm accounts.

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2. Detailed Database Table Schemas

2.1 `users` Table Schema

Column Name	Data Type	Constraint	Field Rules & Description
id	VARCHAR(255)	PRIMARY KEY	Firebase Auth UID / Unique User Identifier
display_name	VARCHAR(255)	NOT NULL	Full name of user or vendor
email	VARCHAR(255)	UNIQUE NOT NULL	Primary email address
phone	VARCHAR(50)	NULLABLE	Contact phone number
role	VARCHAR(50)	NOT NULL	Enum: customer vendor delivery admin
addresses	JSONB	DEFAULT []	Array of address objects with geocodes
shops	JSONB	DEFAULT []	Array of vendor shop objects
created_at	TIMESTAMPTZ	DEFAULT NOW()	Account creation timestamp

2.2 `farms` Table Schema (Agritourism Listings)

Column Name	Data Type	Constraint	Field Rules & Description
id	VARCHAR(255)	PRIMARY KEY	Unique farm ID / Slug (e.g. strawberry-paradise)
farm_name	VARCHAR(255)	NOT NULL	Display title of the farm listing
location	TEXT	NOT NULL	Address & GPS location
description	TEXT	NULLABLE	Farm details narrative
vendor_id	VARCHAR(255)	FK -> users(id)	Owner user ID of vendor
vendor_name	VARCHAR(255)	NULLABLE	Vendor display name
cost_per_person	NUMERIC(10,2)	DEFAULT 0	Ticket fee per visitor (0 = Free Entry)
cost_type	VARCHAR(50)	DEFAULT free	Enum: free payable
accommodation_price	NUMERIC(10,2)	DEFAULT 0	Base price per night for stay options
accommodations	JSONB	DEFAULT []	Array of stay option objects
crops	JSONB	DEFAULT []	Array of crop strings (e.g. ["Spinach"])
fruits	JSONB	DEFAULT []	Array of fruit strings (e.g. ["Mangoes"])
livestock	JSONB	DEFAULT []	Array of animal strings (e.g. ["Gir Cows"])
kids_activities	JSONB	DEFAULT []	Array of kids zone activity strings
farm_products	JSONB	DEFAULT []	Direct harvest product objects
crop_photos	JSONB	DEFAULT []	Produce field photo objects
livestock_photos	JSONB	DEFAULT []	Livestock photo objects
kids_photos	JSONB	DEFAULT []	Kids play photo objects
accommodation_photos	JSONB	DEFAULT []	Stay accommodation photo objects
gallery	JSONB	DEFAULT []	General farm gallery photos

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2.3 `farm\_bookings` Table Schema

Column Name	Data Type	Constraint	Field Rules & Description
id	VARCHAR(255)	PRIMARY KEY	Unique booking ID (e.g. booking_178523...)
farm_id	VARCHAR(255)	FK -> farms(id)	Target farm listing ID
customer_id	VARCHAR(255)	FK -> users(id)	Booking customer user ID
date	VARCHAR(50)	NOT NULL	Scheduled visit date (YYYY-MM-DD)
visitors_count	INTEGER	NOT NULL DEFAULT 1	Number of guests visiting
cost_per_person	NUMERIC(10,2)	DEFAULT 0	Ticket price rate applied
include_stay	BOOLEAN	DEFAULT false	Flag if stay option added
accommodation_title	VARCHAR(255)	NULLABLE	Title of selected accommodation
rooms_booked	INTEGER	DEFAULT 0	Rooms / huts count
stay_cost	NUMERIC(10,2)	DEFAULT 0	Accommodation total cost
total_amount	NUMERIC(10,2)	NOT NULL	Final payable amount in INR
status	VARCHAR(50)	DEFAULT confirmed	Enum: pending confirmed completed cancelled
payment_method	VARCHAR(50)	DEFAULT Online UPI	Payment mode used

2.4 `products` Table Schema (Produce Catalog)

Column Name	Data Type	Constraint	Field Rules & Description
id	VARCHAR(255)	PRIMARY KEY	Unique product ID (e.g. fp-101)
name	VARCHAR(255)	NOT NULL	Produce name (e.g. Organic Strawberries)
category	VARCHAR(100)	NOT NULL	Category: Vegetables Fruits Dairy & Eggs
price	NUMERIC(10,2)	NOT NULL	Sale price per unit in INR (¹)
unit	VARCHAR(50)	DEFAULT 1 kg	Measurement unit (kg, pack, crate)
image	TEXT	NOT NULL	Product photo URL
farm_id	VARCHAR(255)	FK -> farms(id)	Origin farm listing ID
is_deliverable	BOOLEAN	DEFAULT true	False if product is farm pickup only
fulfillment_type	VARCHAR(50)	DEFAULT deliverable	Enum: deliverable non_deliverable pickup_only

3. Sample Payload Contracts

Payload Schema Contract: Agritourism Farm Payload Contract

```
{
  "id": "farm_1785821264",
  "name": "Mannila Organic Paradise & Berry Farm",
  "location": "Mahabaleshwar Highway, Maharashtra 412806",
  "costPerPerson": 250,
  "costType": "payable",
  "crops": ["Strawberries", "Cherry Tomatoes", "Sweet Corn"],
  "fruits": ["Alfonso Mangoes", "Guavas"],
  "livestock": ["Gir Cows", "Free-Range Poultry"],
  "kidsActivities": ["Pottery Workshop", "Petting Zoo"],
  "accommodations": [
    {
      "id": "acc-1",
      "title": "Rustic Mud Hut Stay",
      "price": " ₹1,500 / night",
      "roomQuantity": "2 Huts",
      "roomCapacity": "4 Persons"
    }
  ]
}
```


"farmProducts": [

"id": "fp-101",

"name": "Fresh Organic Strawberries",

"price": 240,

"unit": "500 g box",

"isDeliverable": true

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4. API Mapping & Migration Guidelines

REST API Endpoint	Frontend Usage Contract & Business Function
GET /api/farms	Fetch farm listings with crop/location search filters
GET /api/farms/:id	Fetch complete farm details, crops, stays, products & gallery
POST /api/farms	Create or update farm listing & all sections (Upsert mode)
POST /api/farms/book	Place a farm visit tour or accommodation booking
GET /api/farms/bookings/all	Fetch vendor farm bookings for management dashboard
GET /api/products	Fetch marketplace harvest products by category/farm
POST /api/orders	Place direct harvest marketplace orders

4.1 Engineering & DB Migration Directives

Architecture Directive	Technical Implementation Directive
GIN Indexing	Create GIN indexes on `farms.crops`, `farms.fruits`, and `farms.farm_products` for high-speed tag searching.
Idempotent Upserts	Use `ON CONFLICT (id) DO UPDATE` for `POST /api/farms` to allow progressive saving without duplicate creation.
Currency Precision	Use NUMERIC(10, 2) for monetary values (`cost_per_person`, `total_amount`, `price`) to prevent rounding issues.

